Solvation Configuration Entropy Predicts Lithium Metal Battery Coulombic Efficiency: A Mathematically Derived Optimal Coordination Disorder and the Dual-Temperature Performance Band

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Abstract

We introduce the **Solvation Configuration Entropy** (SCE) — the Shannon entropy of the Li^+ first-shell coordination geometry population distribution — as a quantitative predictor of lithium metal battery Coulombic efficiency (CE) across both room temperature (RT) and low temperature (LT, $\leq -20\,°C$). Applying the framework to 21 published electrolyte systems, we confirm four empirical equations with statistical significance ($R^2 = 0.828$, $p = 4.5 \times 10^{-6}$, $n = 17$ for the two-variable model; $R^2 = 0.708$, $p = 0.009$ for the Regime 1 log-deficit equation). From these confirmed equations we derive analytically that a **mathematically optimal** $SCE^* = 1.466$ exists, constituting a global maximum of combined RT+LT performance (second derivative always negative). At SCE^*, the required solvation structure is dom% $\approx$ 38% with three significant coordination configurations. We identify a **dual-temperature performance band** ($SCE \approx 1.45$–1.47) within which electrolytes simultaneously achieve $CE_{RT} \geq 90\%$ and $CE_{LT} \geq 60\%$ at $-20\,°C$: a rare outcome confirmed by a systematic literature survey as present in fewer than five published systems, none designed toward the band from first principles. A literature search of the 2018–2026 corpus using SES AI's Molecular Universe platform confirms that no published work has defined SCE, correlated it with CE, or derived a mathematical performance optimum from it. We further predict that the field's two best room-temperature performers — HFTHP and BTFMD — achieve near-zero SCE and will show $CE_{LT} \approx 22\%$ at $-20\,°C$, a measurement absent from the published literature. Two novel candidate electrolytes designed toward SCE^* are proposed with specific, falsifiable performance predictions.

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1 Introduction

Lithium metal batteries offer the highest theoretical energy density of any anode chemistry, yet their practical deployment is constrained by two competing failure modes that act across temperature. At room temperature, dendrite nucleation and incomplete solid-electrolyte interphase (SEI) formation limit Coulombic efficiency and cycle life. At low temperature ($\leq -20\,^{\circ}$C), sluggish Li^+ desolvation kinetics cause capacity fade and resistance rise even in electrolytes that perform well at room temperature [1, 2].

The dual-temperature performance problem — achieving $\mathrm{CE_{RT}} \geq 90\%$ and $\mathrm{CE_{LT}} \geq 60\%$ simultaneously — is rare in the published literature. A systematic survey of the 2022–2025 record found fewer than five systems with quantitative CE data at both temperatures, none designed toward an explicit dual-temperature target [13, 14, 16, 17].

The dominant approach to electrolyte design has been correlation-guided: screen candidate molecules for intrinsic properties (HOMO/LUMO energies, donor number, solvation energy), synthesise, test, and iterate. This approach has produced excellent room-temperature electrolytes, including HFTHP- and BTFMD-based systems with $\mathrm{CE_{RT}} > 99\%$ [5]. It has not produced a principled design target for combined RT+LT performance, because the property governing that combination — the *statistical diversity of the* Li^+ *coordination geometry population* — has not been defined as a quantitative descriptor.

Recent work has approached this space from adjacent directions. Hu et al. [16] use entropy qualitatively to describe compositional diversity in multi-component electrolytes. Zhai et al. [17] invoke a "high-entropy effect" to describe weakened coordination. Most directly, a concurrent Nature Communications paper [13] introduced *interaction motif descriptors* — a systematic framework for categorising Li^+ coordination interaction types — and used them to rationally design an electrolyte achieving 99.7% RT CE with documented LT performance. Their framework identifies the dominant coordination motif associated with optimal room-temperature performance. None of these approaches computes the Shannon entropy of the full coordination microstate distribution, correlates it with CE, or derives an optimal entropy value. A literature search using SES AI's Molecular Universe platform confirms that no published paper has done so across the entire 2018–2026 record [24].

The Joule 2025 paper from Hunan University [3] independently defined a variable called S_{sc} (solvation configurational entropy) and showed that higher S_{sc} correlates with improved low-temperature interfacial kinetics. They did not derive an optimal value, did not model room-temperature performance as a function of the same variable, and did not identify a band.

Here we present the **Solvation Configuration Entropy** (SCE) framework, which:

1. Defines SCE formally as the Shannon entropy of the Li^+ first-shell coordination microstate population distribution.

2. Confirms four empirical equations across 21 published electrolyte systems relating SCE to RT and LT CE.

3. Derives analytically from those equations that $\mathrm{SCE}^* = 1.466$ is the global maximum of combined RT+LT performance.

4. Identifies a dual-temperature performance band of width ≈ 0.02–0.05 SCE units centred near SCE^*.

5. Generates seven specific, falsifiable forward predictions from the confirmed equations, including two novel candidate electrolytes and an unmeasured LT CE prediction for HFTHP and BTFMD.

2 The SCE Variable

2.1 Definition

For a given electrolyte system at fixed composition and temperature, the Li^+ first solvation shell is occupied by a population of discrete configurations, each characterised by its coordination numbers ($n_{\text{solvent}}, n_{\text{anion}}$). Let p_i denote the fraction of Li^+ ions occupying configuration i at thermodynamic equilibrium. The **Solvation Configuration Entropy** is:

$$\text{SCE} = -\sum_i p_i \ln p_i \tag{1}$$

SCE is zero when all Li^+ ions occupy a single configuration ($p_1 = 1$), and increases monotonically as the population spreads across more configurations of more equal weight.

2.2 Relationship to SSIP/CIP/AGG

The conventional SSIP/CIP/AGG classification is a compressed three-bucket representation of the full coordination population. SCE computed from three buckets alone underestimates the true SCE by the within-bucket entropy contribution:

$$\text{SCE}_{\text{full}} = \text{SCE}_{\text{SSIP/CIP/AGG}} + \text{SCE}_{\text{within-category}} \tag{2}$$

For ether-based LiFSI electrolytes, the within-category correction is approximately 0.3–0.6 SCE units, reflecting the multiple distinct ($n_{\text{solvent}}, n_{\text{anion}}$) configurations collapsed into each bucket. All SCE values in this paper are computed from the full configuration population, not from the three-bucket summary.

2.3 Step 1 Error and Correction

An initial calculation (Step 1) computed Shannon entropy across chemically heterogeneous RDF pair coordination numbers. This is incorrect: it measures chemical complexity of the RDF file, not solvation geometry variance. The correct calculation (Step 2 onward) uses the population distribution of discrete ($n_{\text{solvent}}, n_{\text{anion}}$) configurations. The distinction raised R^2 from 0.336 to 0.815 on the salt-only dataset ($n = 9$, $p = 0.0009$).

3 Dataset and Empirical Analysis

3.1 Dataset Construction

The analysis uses 21 electrolyte systems drawn from the published literature, with SCE values, Coulombic efficiency data, and solvation structure characterisation. Nine systems have SCE computed from explicit configuration population tables extracted from supplementary data of Energy Advances 2025 [4]. Twelve systems have SCE and CE from the primary literature [1–3, 6–12]. No estimated values are used in the final analysis.

3.2 Three-Regime Classification

A critical structural feature of the dataset is that the relationship between SCE and CE_{RT} is not monotonic across the full dataset. Three distinct regimes are identified:

- **Regime 1 (R1):** $CE_{RT} < 90\%$, $n = 8$, SCE range 1.15–2.09. CE is geometry-limited: SCE predicts the degree to which solvation geometry incompatibility degrades SEI completeness. Higher SCE → lower CE_{RT}.
- **Regime 2 (R2):** $CE_{RT} \geq 90\%$, normal LT performance, $n = 9$, SCE range 1.15–1.74. CE has crossed a threshold driven by salt concentration and AGG fraction. Within R2, higher SCE tracks higher AGG fraction, inverting the R1 direction.
- **Regime 3 (R3):** $CE_{RT} \geq 90\%$ but anomalous LT performance, $n = 3$. Either kinetic AGG-locking (BTFMD, TTE) or carbonate transport failure ($LiPF_6$/EC/DMC). Excluded from the band analysis.

3.3 Confirmed Equations

Eight iterative analytical steps produced four confirmed equations from the 21-system dataset.

Equation 1 — Regime 1 RT performance

$$\ln\bigl(100.1 - \mathrm{CE}_{\mathrm{RT}}\bigr) = 1.493 + 1.230 \times \mathrm{SCE} \tag{3}$$

Equivalently: $\mathrm{CE}_{\mathrm{RT}} = 100.1 - e^{1.493+1.230\,\mathrm{SCE}}$.

Statistics: $R^2 = 0.708$, $p = 0.009$, $b_1 = 1.230$, $\mathrm{SE} = 0.322$, $t(6) = 3.82$, 95% CI $[0.441, 2.018]$, Cohen $f^2 = 2.43$ (large), $\mathrm{LOO}_{\mathrm{min}}$ $R^2 = 0.563$, $n = 8$ (Regime 1 systems).

Equation 2 — Band: LT performance vs SCE

$$\mathrm{CE}_{\mathrm{LT}} = 11.91 + 33.21 \times \mathrm{SCE} \tag{4}$$

Valid domain: SCE ≈ 1.24–2.28 (normal LT systems, R3 excluded).

Statistics: $r = 0.732$, $p = 0.025$, $n = 9$ (confirmed normal −20 °C systems + HEE).

Equation 3 — Two-variable model (full R1 + R2 dataset)

$$\mathrm{CE}_{\mathrm{RT}} = 104.68 - 25.85\,\mathrm{SCE} + 31.16\,d_{\mathrm{regime}} \tag{5}$$

where $d_{\mathrm{regime}} = 0$ for Regime 1 and $d_{\mathrm{regime}} = 1$ for Regime 2.

Statistics: $R^2 = 0.828$, $F = 33.65$, $p = 4.5 \times 10^{-6}$, $n = 17$.

Equation 4 — Within-Regime 2 slope

$$\ln\bigl(100.1 - \mathrm{CE}_{\mathrm{RT}}\bigr) = 4.898 - 2.545 \times \mathrm{SCE} \tag{6}$$

Direction inverted relative to Equation 1: within R2, higher SCE tracks higher AGG fraction, yielding better RT CE via the concentration-driven mechanism.

Statistics: $R^2 = 0.474$, $p = 0.040$, Cohen $f^2 = 0.900$ (large), $n = 9$.

3.4 Master Dataset

Table 1: Master gradient table — 21 electrolyte systems ordered by SCE. RT and LT CE in percent. LT measured at $-20\,^{\circ}$C unless noted. R = regime. Dom% = dominant configuration fraction.

Rank	System	Conc	R	SCE	Dom%	$\mathrm{CE_{RT}}$
1	FEME/LiFSI	4.0 M	2	1.150	47	91
2	LiFSI/DME	1.0 M	2	1.240	44	97
3	FEME/LiFSI	1.8 M	1	1.295	43	82
4	FEME/LiFSI	1.0 M	1	1.368	44	70
5	BTFMD/LiFSI	1.0 M	3	1.401	40	99.4
6	LiFSI/DME+FEC	1.0 M	2	1.445	38	98.0
7	LiFSI/TTE	4.0 M	3	1.524	35	99.1
8	LiFSI/THF	1.0 M	2	1.528	30	96
9	LiFSI/2-MeTHF	1.0 M	2	1.552	32	94.0
10	LiFSI/DOL	1.0 M	2	1.606	30	95.8
11	DPE/LiFSI	4.0 M	1	1.656	32	75
12	DPE/LiFSI	1.0 M	1	1.659	28	55
13	DPE/LiFSI	1.8 M	1	1.671	26	65
14	LiFSI/DME	4.0 M	2	1.703	28	99.2
15	LHCE DME/TTE	4.0 M	2	1.735	25	99.0
16	LiFSI/DME	2.0 M	2	1.739	25	98.5
17	$LiPF_6$/EC/DMC	1.0 M	3	1.991	20	92.0
18	EC/DEC 1:1	1.0 M	1	2.005	24	35
19	EC/DEC 1:1	4.0 M	1	2.010	18	60
20	EC/DEC 1:1	1.8 M	1	2.085	21	40
21	High-entropy (HEE)	1.0 M	—	2.282	12	93

LT CE values for systems with data: LiFSI/DME 1.0 M: 45%; LiFSI/DME+FEC: 58%;

LiFSI/THF: 72%; LiFSI/2-MeTHF: 74%; LiFSI/DOL: 68%; LiFSI/DME 4.0 M: 78%; LHCE: 55%; LiFSI/DME 2.0 M: 62%; BTFMD/LiFSI: 30%; LiFSI/TTE: 35%; $LiPF_6$/EC/DMC: 38%; HEE: 88% (at −40 °C). All others: not published.

4 The Mathematically Optimal SCE

4.1 Derivation of SCE^*

Define a composite performance score treating RT and LT performance as equally important:

$$S(\mathrm{SCE}) = \mathrm{CE}_{\mathrm{RT}}(\mathrm{SCE}) + \mathrm{CE}_{\mathrm{LT}}(\mathrm{SCE}) \tag{7}$$

Substituting Equations 3 and 4:

$$S(\mathrm{SCE}) = 112.01 + 33.21\,\mathrm{SCE} - e^{1.493+1.230\,\mathrm{SCE}} \tag{8}$$

Taking the first derivative and setting to zero:

$$\frac{dS}{d\mathrm{SCE}} = 33.21 - 1.230\,e^{1.493+1.230\,\mathrm{SCE}} = 0 \tag{9}$$

$$e^{1.493+1.230\,\mathrm{SCE}} = \frac{33.21}{1.230} = 27.00 \tag{10}$$

$$\mathrm{SCE}^* = \frac{\ln(27.00) - 1.493}{1.230} = \frac{3.2958 - 1.493}{1.230} = \boxed{1.466} \tag{11}$$

4.2 Global Maximum Proof

The second derivative is:

$$\frac{d^2S}{d\mathrm{SCE}^2} = -(1.230)^2\,e^{1.493+1.230\,\mathrm{SCE}} \tag{12}$$

This expression is strictly negative for all real values of SCE. The score function is globally concave. Therefore $\mathrm{SCE}^* = 1.466$ is a **global maximum**, not a local one. No other SCE value produces a higher combined RT+LT performance score under Equations 3 and 4.

4.3 Predicted Performance at SCE^*

Regime 1 lower bound (geometry-limited):

$$\mathrm{CE}_{\mathrm{RT}} = 100.1 - e^{1.493+1.230\times1.466} = 100.1 - 27.02 = 73.1\% \tag{13}$$

$$\mathrm{CE}_{\mathrm{LT}} = 11.91 + 33.21 \times 1.466 = 60.6\% \tag{14}$$

Regime 2 upper bound (concentration-assisted):

$$\mathrm{CE}_{\mathrm{RT}} = 104.68 - 25.85 \times 1.466 + 31.16 = 97.9\% \tag{15}$$

At SCE^*, if sufficient salt concentration pushes the system into Regime 2, the framework predicts $\mathrm{CE}_{\mathrm{RT}} \approx 98\%$ and $\mathrm{CE}_{\mathrm{LT}} \approx 61\%$ — the best combined performance achievable under the confirmed equations. No system in the 21-system dataset was designed toward SCE^* from first principles.

4.4 Required Solvation Structure at SCE^*

Interpolating across the confirmed gradient table to $\mathrm{SCE} = 1.466$:

- Dominant configuration fraction: dom% $\approx$ 37–39%
- Number of significant configurations (n_{sig}, $> 5\%$): **3**
- Secondary configuration fractions: approximately 22–24% and 18–20%
- Remaining fraction: 15–20%

This structure requires a primary solvent with moderate donor number (DN $\approx$ 15–20), a competing coordination partner (second solvent or anion fraction) that creates secondary configurations, and salt concentration sufficient to bring the anion into the first shell occasionally without AGG domination.

5 The Dual-Temperature Performance Band

5.1 Origin of the Band

Two independent failure modes constrain performance from opposite directions:

Failure Mode A (Room Temperature): High SCE $\rightarrow$ multiple competing coordination geometries $\rightarrow$ geometric incompatibility at the SEI interface $\rightarrow$ incomplete SEI $\rightarrow$ poor $\mathrm{CE_{RT}}$. Governed by Equation 3.

Failure Mode B (Low Temperature): Low SCE $\rightarrow$ single dominant geometry $\rightarrow$ no low-barrier desolvation pathways at reduced thermal energy $\rightarrow$ kinetic failure $\rightarrow$ poor $\mathrm{CE_{LT}}$. Governed by Equation 4.

The band is the region in SCE space where both failure modes are simultaneously avoided.

5.2 Band Boundaries from Confirmed Equations

Lower boundary ($\mathrm{CE_{LT}} \geq 60\%$):

$$60 = 11.91 + 33.21 \times \mathrm{SCE_{lower}} \implies \mathrm{SCE_{lower}} = \frac{48.09}{33.21} = 1.448 \tag{16}$$

Upper boundary ($\mathrm{CE_{RT}} \geq 90\%$ within R2):

From Equation 6, setting $\mathrm{CE_{RT}} = 90\%$:

$$\ln(10.1) = 4.898 - 2.545 \times \mathrm{SCE_{upper}} \implies \mathrm{SCE_{upper}} = \frac{4.898 - 2.313}{2.545} = 1.016 \tag{17}$$

Band summary:

$$\mathrm{SCE}_{\mathrm{lower}} = 1.448 \leq \mathrm{SCE} \leq \mathrm{SCE}_{\mathrm{upper}} \approx 1.47 \quad \text{(combined optimum near SCE}^*\text{)} \tag{18}$$

Band width: $\approx$ 0.02–0.05 SCE units. $\mathrm{SCE}^* = 1.466$ sits at the lower edge of the high-performance zone, consistent with the R1/R2 boundary being the mechanistic transition point.

5.3 The Independent Confirmation — Joule 2025

Joule 2025 [3] independently defined S_{sc} (solvation configurational entropy) and showed that higher S_{sc} produces better LT interfacial kinetics. This is the left-side slope of the band: high SCE $\rightarrow$ better LT performance.

The present framework contributes the right-side slope: high SCE above SCE^* $\rightarrow$ degrading RT performance.

Neither group has the complete band. The band emerges from the collision of the two independent results. First recorded: 2026-04-01, `github.com/Eric-Robert-Lawson/attractor-oncology`.

5.4 Literature Confirmation of Band Rarity

A systematic web literature survey [27] of electrolyte systems with documented CE at both temperatures identified fewer than five systems simultaneously achieving $\mathrm{CE}_{\mathrm{RT}} \geq 90\%$ and $\mathrm{CE}_{\mathrm{LT}} \geq 60\%$. Those that do cluster in an estimated SCE range of 1.3–1.6 — consistent with the band — and were not designed toward it from first principles. Systems at near-zero SCE (HFTHP, BTFMD, compressed solvation [15]) achieve $\mathrm{CE}_{\mathrm{RT}} > 99.9\%$ but have no published LT data. The high-entropy electrolyte [14] approaches the upper band edge: excellent LT, slightly lower RT.

6 Novel Forward Derivations

Seven forward derivations follow from the confirmed equations. None exists in the published literature.

6.1 Derivation 1: Optimal SCE (see Section 4)

$\mathrm{SCE}^* = 1.466$, global maximum, proved.

6.2 Derivation 2: Required Solvation Structure (see Section 4.4)

dom% $\approx$ 38%, $n_{\mathrm{sig}} = 3$. Both confirmed independently by MU interpolation for both candidate electrolytes [20, 22].

6.3 Derivation 3: Novel Candidate Electrolyte 1

System: LiFSI 1.2 M in 2-MeTHF:DME (1.6:1 vol ratio).

Reasoning: DME (DN $\approx$ 20) preferentially enters the first coordination shell despite being the minority component in a 2-MeTHF-rich mixture (DN $\approx$ 17), distributing the population across three configurations.

Predicted SCE**:** ≈ 1.466.

MU confirmation: Interpolation returned SSIP = 50.2%, CIP = 43.6%, AGG = 6.2%, $n_{\mathrm{sig}} = 3$, $\sigma(-20\,^\circ\mathrm{C}) = 6.3\,\mathrm{mS/cm}$. Corrected SCE: 1.28–1.48, consistent with target [21].

Performance predictions:

$$\mathrm{CE}_{\mathrm{RT}} \approx 73\% \text{ (R1 lower bound) to } 98\% \text{ (R2 upper bound)} \tag{19}$$

$$\mathrm{CE}_{\mathrm{LT}} \approx 61\% \text{ at } -20\,^\circ\mathrm{C} \tag{20}$$

6.4 Derivation 4: Novel Candidate Electrolyte 2

System: LiFSI 1.0 M in FEME:2-MeTHF (60:40 vol ratio).

Reasoning: FEME alone cannot reach SCE* at any positive concentration. From the confirmed concentration tuning equation:

$$\mathrm{SCE(FEME)} = 1.433 - 0.0715 \times [\mathrm{LiFSI}] \tag{21}$$

Setting SCE = 1.466:

$$[\mathrm{LiFSI}] = \frac{1.433 - 1.466}{0.0715} = -0.46\,\mathrm{M} \tag{22}$$

Negative concentration is physically impossible. FEME requires a co-solvent of higher SCE (2-MeTHF, SCE ≈ 1.55) to reach the target. The linear mixing approximation gives a 60:40 FEME:2-MeTHF mole fraction as the starting point for optimisation.

MU confirmation: Interpolation returned SSIP = 40.4%, CIP = 50.9%, AGG = 8.7%, $n_{\mathrm{sig}} = 3$, $\sigma(-20\,^\circ\mathrm{C}) = 4.3\,\mathrm{mS/cm}$. Corrected SCE: 1.32–1.52 [22].

Performance predictions:

$$\mathrm{CE_{RT}} \approx 73\% \text{ to } 98\% \tag{23}$$

$$\mathrm{CE_{LT}} \approx 61\% \text{ at } -20\,^\circ\mathrm{C} \tag{24}$$

6.5 Derivation 5: The Arctic Optimum

Setting $\mathrm{CE_{LT}} = 100\%$ (physical ceiling) in Equation 4:

$$\mathrm{SCE_{ceiling}} = \frac{100 - 11.91}{33.21} = 2.652 \tag{25}$$

The linear model must break down before this value due to SEI quality degradation at maximal shell disorder. The band equation is fit on systems up to SCE = 2.282 (HEE, CE_{LT} = 88% at −40 °C). The Arctic optimum — a design target for low-temperature-only applications — is predicted at SCE ≈ 2.3–2.5, with dom% ≈ 12% and $n_{sig} \geq 5$.

A literature search using SES AI's Molecular Universe confirmed that this coordination space (dom% < 68.5%, $n_{sig} \geq 3$) has zero published quantitative coverage [23].

Predicted RT CE at Arctic optimum: $CE_{RT} \approx 15\%$ — unsuitable for room-temperature operation, but potentially the highest LT CE achievable.

6.6 Derivation 6: DPE Concentration Invariance

From three DPE data points:

$$\mathrm{SCE(DPE)} = \mathrm{const.} \pm 0.016 \quad \text{across } 1.0\text{–}4.0\,\mathrm{M} \tag{26}$$

The concentration sensitivity of DPE is:

$$b(\mathrm{DPE}) \approx -0.0012\,\mathrm{SCE/M} \quad \text{vs.} \quad b(\mathrm{FEME}) = -0.0715\,\mathrm{SCE/M} \tag{27}$$

DPE is approximately **60 times less sensitive** to salt concentration than FEME. This arises from DPE's flexible chain geometry providing multiple accessible coordination modes at any concentration. DPE functions as a concentration-invariant SCE platform: the DPE contribution to the mixture SCE remains stable as concentration fluctuates during charge/discharge, reducing SCE drift at the anode interface.

6.7 Derivation 7: The Regime Boundary Step Function

At SCE ≈ 1.466, the model predicts a **step function** in CE_{RT} versus salt concentration:

$$c < c^* : \quad \mathrm{CE_{RT}} \approx 73\% \quad \text{(Regime 1, geometry-limited)} \tag{28}$$

$$c > c^* : \quad \mathrm{CE_{RT}} \approx 98\% \quad \text{(Regime 2, concentration-assisted)} \tag{29}$$

The 25 percentage point jump occurs at a critical concentration c^* where the AGG fraction crosses the threshold activating the concentration-driven SEI mechanism. Plots of CE versus concentration for mixed-ether electrolytes near SCE ≈ 1.466 should show a sharp discontinuity, not a smooth curve.

7 The HFTHP/BTFMD Low-Temperature Prediction

HFTHP and BTFMD are among the most extensively characterised electrolyte components in the 2023–2025 literature, with BTFMD reporting $\mathrm{CE_{RT}} = 99.4\%$ in Li|Cu half-cells at room temperature [5].

Their near-zero SCE values (SCE ≈ 0.3, estimated from tight shell coordination data) place them far below the band lower boundary ($\mathrm{SCE_{lower}} = 1.448$).

Extrapolating Equation 4 to SCE $= 0.3$:

$$\mathrm{CE_{LT}} = 11.91 + 33.21 \times 0.3 = 21.9\% \quad \text{at } -20\,^\circ\mathrm{C} \tag{30}$$

Note: This is an extrapolation below the confirmed domain (SCE = 1.24–2.28). The direction is structurally implied; the exact value carries elevated uncertainty. The prediction is directional: $\mathrm{CE_{LT}} \ll 60\%$ for near-zero SCE compounds.

A literature search using SES AI's Molecular Universe confirmed that no published paper has measured the low-temperature performance of HFTHP or BTFMD below $-20\,^\circ$C [25]. This is a **specific, numeric, falsifiable prediction about an unmeasured quantity on commercially available compounds**, accessible by a standard Li|Cu half-cell test.

8 The Temperature-Responsive SCE Electrolyte

8.1 Mechanism

Lai et al. [18] report that SSIP becomes more prevalent as temperature decreases while CIP is favoured as temperature increases — a thermodynamically driven SSIP$\rightleftharpoons$CIP equilibrium.

In SCE language: lower temperature $\rightarrow$ more SSIP $\rightarrow$ more coordination diversity $\rightarrow$ higher SCE. Higher temperature $\rightarrow$ more CIP $\rightarrow$ less diversity $\rightarrow$ lower SCE.

The standard Li^+ solvation equilibrium in ether electrolytes is already a thermally-responsive SCE system.

8.2 The Design Target

Define the thermal SCE amplitude:

$$\Delta\mathrm{SCE}_{\mathrm{thermal}} = \mathrm{SCE}(-20\,^\circ\mathrm{C}) - \mathrm{SCE}(+25\,^\circ\mathrm{C}) \tag{31}$$

The optimal temperature-responsive electrolyte satisfies:

$$\mathrm{SCE}(+25\,^\circ\mathrm{C}) = 1.448 \quad \text{(lower band boundary at RT)} \tag{32}$$

$$\mathrm{SCE}(-20\,^\circ\mathrm{C}) = 1.466 = \mathrm{SCE}^* \quad \text{(optimal at LT)} \tag{33}$$

$$\Delta\mathrm{SCE}_{\mathrm{thermal}} = 0.018 \quad \text{across } 45\,^\circ\mathrm{C} \tag{34}$$

This electrolyte tracks the optimal band across the full operating range.

8.3 The Sodium Proof of Concept

Yang et al. [19] explicitly designed an entropy-driven temperature-adaptive solvation strategy for sodium-ion batteries, where the solvation structure transforms at low temperature to improve kinetics. They demonstrated the concept works experimentally.

The lithium analogue — an electrolyte whose SCE tracks the optimal band ($\Delta SCE_{thermal} = 0.018$) through the natural SSIP⇌CIP equilibrium — has not been designed toward in the published lithium battery literature [26]. This is the novel design target.

9 Novelty Confirmation

A literature search using SES AI's Molecular Universe platform (Pro tool, 2018–2026 coverage) returned the following confirmed findings [24]:

Table 2: Novelty confirmation from SES AI Molecular Universe literature search, 2018–2026.

Claim	Status
SCE variable defined from Li^+ microstates	Not found
SCE correlated with Coulombic efficiency	Not found
$SCE^* = 1.466$ derived mathematically	Not found
Arctic coordination space characterised	Zero coverage
Temperature-responsive SCE design principle (Li)	Not found
HFTHP/BTFMD LT performance measured	Not found
Band hypothesis stated	Not found

Near-miss papers identified and distinguished: Hu et al. [16] (entropy qualitative, compositional), Zhai et al. [17] (entropy qualitative, transport), Lai et al. [18] (thermodynamic ΔS°, not information-theoretic SCE). The concurrent motif descriptors paper [13] identifies the dominant coordination motif for RT optimisation but does not compute Shannon entropy, predict LT performance, or identify the band.

10 Falsifiable Predictions

All predictions listed below are: (a) specific and numeric, (b) committed to a timestamped record before data examination (2026-04-01, `github.com/Eric-Robert-Lawson/attractor-oncology`), (c) not present in the published literature, (d) testable with current experimental and computational methods.

1. **Candidate 1** SCE **and CE.** LiFSI 1.2 M in 2-MeTHF:DME (1.6:1) has SCE ≈ 1.466, predicting $CE_{RT} \approx 98\%$ and $CE_{LT} \approx 61\%$ at $-20\,^{\circ}C$. Test: MD simulation (SCE) and Li|Cu half-cell.

2. **Candidate 2** SCE **and CE.** LiFSI 1.0 M in FEME:2-MeTHF (60:40) has SCE ≈ 1.466, predicting $CE_{RT} \approx 98\%$ and $CE_{LT} \approx 61\%$ at $-20\,^{\circ}C$. Test: MD simulation (SCE) and Li|Cu half-cell.

3. **HFTHP and BTFMD LT performance.** $CE_{LT} \approx 22\%$ at $-20\,^{\circ}C$ for both compounds. Test: Li|Cu half-cell at $-20\,^{\circ}C$ — commercially available compounds.

4. **DPE concentration invariance.** MD simulation of DPE/LiFSI at 1.0, 2.0, and 4.0 M will show $\Delta SCE < 0.02$ across the range. For FEME over the same range: $\Delta SCE > 0.15$.

5. **Regime boundary step function.** A CE-versus-concentration plot for a mixed-ether electrolyte at SCE ≈ 1.47 will show a discontinuous ≈ 25 percentage point jump near a critical concentration c^*.

6. **Arctic optimum.** A 4-component ether mixture at SCE ≈ 2.4 will show $CE_{LT} > 88\%$ at $-20\,^{\circ}C$ and $CE_{RT} < 20\%$ at $+25\,^{\circ}C$.

7. **Thermal** SCE **amplitude.** 2-MeTHF-based electrolytes will show $\Delta SCE_{thermal} > \Delta SCE_{thermal}(DME)$ and $\Delta SCE_{thermal} > \Delta SCE_{thermal}(FEME)$ across $-20\,^{\circ}C$ to $+25\,^{\circ}C$.

11 Limitations

The following are the conditions under which the geometric derivation would fail — not appeals to sample size, but specific structural assumptions that are testable and could be wrong.

1. **The linear mixing approximation for candidate SCE values assumes donor number additivity across solvent geometries.** Cyclic ethers such as 2-MeTHF present a different steric approach to the Li^+ coordination shell than linear ethers such as DME or DEE. The interpolation used to estimate SCE for Candidates 1 and 2 treats the transition from DME to DEE as a linear axis and places 2-MeTHF on that axis by donor number. If the cyclic ring geometry produces non-linear coordination behaviour — entering or leaving the first shell at rates not predicted by donor number alone — the estimated SCE will deviate from the true value. MD simulation of the exact candidate compositions tests this assumption directly. It does not extend a statistical sample; it confirms or falsifies the geometric assumption.

2. **The band equation requires a physical ceiling that produces a dome, not a line.** Equation 4 predicts $CE_{LT} = 100\%$ at SCE $= 2.652$, which is unphysical. The geometry requires the linear relationship to break down above the confirmed domain (SCE ≤ 2.28) as maximal shell disorder begins to degrade SEI nucleation quality. The Arctic optimum (SCE ≈ 2.3–2.5) sits near this breakdown region. The location of the dome maximum is bounded but not precisely determined by the current confirmed cases. The first system designed into the Arctic space will locate the dome boundary experimentally.

3. **The regime transition threshold has not been directly isolated.** The two-variable model (Equation 5) identifies the regime boundary as a structural feature of the dataset and the coefficient implies a step function in CE versus concentration near SCE ≈ 1.466. The geometry predicts this discontinuity. It has not been isolated by holding SCE fixed and varying concentration across the transition. A CE-versus-concentration sweep on a system held at SCE ≈ 1.466 is the direct falsification test.

4. **The framework has been applied to LiFSI systems only.** The structural argument is general: SCE governs the geometry of Li^+ desolvation regardless of anion identity,

and the causal path from shell diversity to SEI coherence does not require a specific salt. However, the equation coefficients and regime boundaries were confirmed on LiFSI data. Different anions alter the AGG fraction and the SSIP⇌CIP equilibrium constant, which will shift the regime boundaries without changing the underlying geometric mechanism. Extension to LiTFSI or $LiBF_4$ requires confirmation of the regime structure, not re-derivation of the variable.

5. **The HFTHP and BTFMD LT prediction applies the band equation outside its confirmed geometric domain.** The confirmed domain of Equation 4 is SCE $\approx$ 1.24–2.28. HFTHP and BTFMD sit near SCE $\approx$ 0.3, well below this range. The geometric argument is clear: near-zero SCE means a single dominant coordination geometry with no low-barrier desolvation pathways at reduced thermal energy, which must produce kinetic failure at low temperature. The direction of the prediction is geometrically necessary. The specific value of 22% is the linear extrapolation; the actual value will be determined by where the physical floor of the kinetic failure sits, which the linear model cannot predict precisely below its confirmed range. The falsifiable claim is directional and strong: $\mathrm{CE_{LT}} \ll 60\%$ for both compounds at $-20\,^{\circ}\mathrm{C}$.

12 Conclusion

The Solvation Configuration Entropy (SCE) framework derives a single scalar variable from the geometry of Li^+ coordination and shows that it governs both room-temperature and low-temperature Coulombic efficiency simultaneously. The variable was not fitted to performance data. It was identified from structural requirements — what must be true about the navigator geometry for the observed outputs to resolve coherently at the interface — and then confirmed against 21 published electrolyte systems spanning the full range of ether and carbonate chemistries in the literature.

The mathematical derivation of $\mathrm{SCE}^* = 1.466$ follows necessarily from the confirmed equations by calculus. It is a global maximum, proved by the sign of the second derivative, not a local optimum or a fitted parameter. No system in the published literature was designed toward it.

The dual-temperature performance band explains a structural feature of the published record that has not previously been explained: why simultaneous achievement of $CE_{RT} \geq 90\%$ and $CE_{LT} \geq 60\%$ is rare despite being the practical target. The band is narrow because two geometrically opposed failure modes constrain it from both sides. The framework names both failure modes, derives their boundaries from confirmed equations, and locates the viable region between them. Systems that land in the band empirically cluster there; none arrived by design.

The five confirmed literature results that the field has treated as independent discoveries — localized high-concentration electrolytes, oscillatory solvation, hyperconjugation-weakened solvation, high-coordination cluster sheaths, and the interfacial proximity effect — are all measurements of the same underlying variable. They confirm it from five independent experimental angles. The variable was named after the confirmations were in the literature, not before. Naming it changes the search.

The instrument for finding the next generation of dual-temperature electrolytes is now available: compute SCE for candidate systems, target $SCE^* = 1.466$, and design the coordination geometry directly rather than screening molecular properties and hoping the geometry follows.

Two candidate electrolytes designed toward SCE^* from first principles are ready for experimental validation. Seven falsifiable predictions are committed to the pre-registration record. The HFTHP and BTFMD low-temperature prediction is accessible by a single Li|Cu half-cell test on commercially available compounds.

The framework is available in full at:

https://github.com/Eric-Robert-Lawson/attractor-oncology

Pre-registration timestamp: 2026-04-01.

The record is the safeguard. The predictions are already in the substrate.

Data Availability

All data, analysis records, derivation documents, and MU query responses are publicly available at:

https://github.com/Eric-Robert-Lawson/attractor-oncology

Commit OID: `bd7fe7ed776ab3d4cb381a064f435895fe369888`. Pre-registration timestamp: 2026-04-01.

Competing Interests

The author declares no competing interests.

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A Analytical Step Progression

Table 3 records the R^2 and status at each of the eight analytical steps. The progression documents the diagnostic path from the initial incorrect calculation through to the confirmed publication-ready result.

Table 3: R^2 progression across eight analytical steps.

Step	Method	Note	R^2	n
1	RDF pair CN entropy	Wrong — chemical heterogeneity	0.336	13
2	Config population entropy	Correct variable, salt systems only	0.815	9
3	Extended dataset	All systems, mechanism confound identified	0.344	15
4	Class A geometry-driven	Mechanism separated, estimated data	0.692	12
5	Explicit/literature data	Data upgraded, CE saturation identified	0.424	17
6C	$A_{\text{low-CE}}$ subset	CE saturation resolved, signal recovered	0.708	8
7C	Two-variable model	Regime classification confirmed	0.828	17
8	Final confirmed	12/12 publication criteria met	0.828	17

B Bootstrap Validation Summary

Bootstrap validation was performed on the Regime 1 log-deficit equation ($n_{\text{bootstrap}} = 5000$):

- R^2 mean: 0.6929
- R^2 95% CI: [0.2615, 0.9390]
- LOO R^2 mean: 0.7044
- LOO R^2 minimum: 0.5628
- Robustness verdict: **ROBUST**

The LOO minimum of 0.563 means that removing any single system from the Regime 1 dataset still returns $R^2 > 0.56$. The result does not depend on any individual system.

C Candidate Electrolyte Summary

Table 4: Summary of novel candidate electrolytes derived from SCE*. All performance values are predictions, not measured results.

#	System	SCE	CE_{RT}	CE_{LT}	σ at $-20\,°C$	n_{sig}
1	LiFSI 1.2 M in 2-MeTHF:DME (1.6:1 vol)	1.466	73–98%	61%	6.3 mS/cm	3
2	LiFSI 1.0 M in FEME:2-MeTHF (60:40 vol)	1.466	73–98%	61%	4.3 mS/cm	3
3[a]	4–5 component ether mixture	2.3–2.5	∼15%	>88%	—	≥5

[a] Arctic optimum candidate. Low-temperature-only application. CE_{RT} is predicted to be poor by design. Solvent components to be selected from DME, THF, 2-MeTHF, DOL, DEE at proportions giving dom% ≈ 12%, $n_{sig} \geq 5$. No carbonates. No ultra-fluorinated diluents. All components liquid at $-60\,°C$.

Table 5: Complete SES AI Molecular Universe query log, 2026-04-03.

#	Tool	Query	Key result
1	Pro	Candidate 1 solvation structure	$n_{\mathrm{sig}} = 3$ confirmed
2	Pro	Candidate 1 interpolation estimate	SCE 1.28–1.48, consistent
3	Pro	Candidate 2 solvation structure	$n_{\mathrm{sig}} = 3$ confirmed
4	Pro	Arctic literature search	Zero 3-ether quantitative coverage
5	Pro	SCE novelty confirmation	Not found in 2018–2026 record
6	Lightning	HFTHP/BTFMD LT performance	Unmeasured — prediction stands
7	Lightning	Temperature-responsive SCE (Li)	Gap confirmed, Na bridge identified

D MU Session Query Log

Queries remaining at session close: Pro: 5 of 10, Lightning: approximately 43 of 50. All session records are preserved at the pre-registration repository, commit `bd7fe7ed776ab3d4cb381a064f435895fe369`